

Dynamic Guideline: WHO-Based Leishmaniasis - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

Latest WHO/PAHO leishmaniasis treatment guideline identified:

- **Guideline Title:** *Guideline for the Treatment of Leishmaniasis in the Americas. Second Edition*
- **Publication Date:** 2022
- **Issuing Organization:** Pan American Health Organization / World Health Organization (PAHO/WHO)
- **Target Population:** People with cutaneous, mucosal, and visceral leishmaniasis in the Region of the Americas, including special populations such as immunocompromised patients and people with visceral leishmaniasis and HIV coinfection.
- **Scope and Objectives:** To update PAHO's 2013 GRADE-based treatment recommendations using new evidence, and to provide therapeutic indications, treatment criteria, and regimens adapted to the epidemiology, species distribution, safety concerns, and feasibility issues in the Americas [PAHO/WHO guideline](#), [PAHO Digital Library](#), [PAHO news](#).

PAHO/WHO reports that the second edition contains **16 recommendations and 13 good-practice points** covering pharmacologic treatment of **cutaneous, mucosal, and visceral leishmaniasis**, as well as case management and secondary prophylaxis for people with visceral leishmaniasis and HIV or other immunosuppressive conditions [PAHO Journal of Public Health](#), [PMC](#).

Exactly five WHO or PAHO/WHO leishmaniasis guideline / technical guidance documents synthesized

#	WHO / PAHO-WHO document	Date	Main scope	GRADE status
1	<i>Guideline for the Treatment of Leishmaniasis in the Americas. Second Edition</i>	2022	Treatment of cutaneous, mucosal, and visceral leishmaniasis in the Americas	GRADE-based update PAHO Digital Library
2	<i>Guideline for the Treatment of Leishmaniasis in the Americas</i>	2013	Earlier PAHO treatment recommendations for the Americas	Developed using GRADE, later updated in 2022 PAHO Digital Library
3	<i>WHO guideline for the treatment of visceral leishmaniasis in HIV co-infected patients in East Africa and South-East Asia</i>	7 June 2022	VL caused by <i>Leishmania donovani</i> in HIV-coinfected patients in East Africa and South-East Asia	GRADE-based, WHO Handbook process WHO , NCBI Bookshelf
4	<i>Control of the leishmaniases: WHO Technical Report Series 949</i>	2010	Global control, diagnosis, treatment, surveillance, and programme guidance	Expert-committee technical report; GRADE not stated in supplied

#	WHO / PAHO-WHO document	Date	Main scope	GRADE status
				sources WHO TRS 949 PDF
5	<i>Plan of Action to Strengthen Leishmaniasis Surveillance and Control in the Americas 2017–2022</i>	2017–2022	Regional surveillance and control framework for endemic countries in the Americas	Programmatic plan; GRADE not stated PAHO Digital Library

Clarification on “Meaningful youth engagement” WHO title

A WHO publication matching the phrase “**Meaningful youth engagement in a WHO guideline development process**” was identified, but it is **not a leishmaniasis guideline**. The exact identified WHO publication is “**Meaningful youth engagement in a WHO guideline development process: case study of WHO abortion care guideline (2022)**”, dated **11 December 2025** [WHO publication page](#), [WHO event page](#). No supplied source documents youth-specific engagement in WHO leishmaniasis guideline development; the leishmaniasis guideline sources document stakeholder input, equity, acceptability, feasibility, and geographic/gender representation, but not youth-specific participation [NCBI Bookshelf](#).

1.2 Key Recommendations

A. PAHO/WHO 2022 guideline: treatment of leishmaniasis in the Americas

Clinical area	Recommendation	GRADE Evidence Quality	Strength
Cutaneous leishmaniasis	Do not use ketoconazole as a therapeutic option; it was removed from PAHO/WHO options in the second edition.	Low*	Conditional*
Cutaneous leishmaniasis	Use intralesional pentavalent antimonials in appropriate patients with cutaneous leishmaniasis.	Moderate*	Strong
Cutaneous leishmaniasis	Consider miltefosine for species in which evidence supports efficacy; PAHO/WHO reports that strong evidence for miltefosine expanded from two to four <i>Leishmania</i> species.	Moderate*	Conditional*
Mucosal leishmaniasis	Use pentavalent antimonials with or without oral pentoxifylline for mucosal leishmaniasis.	Moderate*	Strong
Visceral leishmaniasis	Use liposomal amphotericin B where indicated and available.	Moderate*	Strong*

Clinical area	Recommendation	GRADE Evidence Quality	Strength
Visceral leishmaniasis	Use pentavalent antimonials only conditionally ; PAHO/WHO reports that recommendations previously considered strong are now conditional.	Low*	Conditional
Visceral leishmaniasis	Use amphotericin B deoxycholate only conditionally ; PAHO/WHO reports that previous strong recommendations are now conditional.	Low*	Conditional
Visceral leishmaniasis caused by <i>Leishmania infantum</i>	Avoid miltefosine ; PAHO/WHO reports strong evidence against its use in this setting.	Moderate*	Strong against
Immunocompromised patients	Avoid pentavalent antimonials in immunocompromised patients.	Low*	Conditional against*
VL with HIV or other immunosuppression	Address case management and consider secondary prophylaxis according to relapse risk and regional feasibility.	Low*	Conditional*

*The PAHO/WHO sources confirm GRADE methodology and report selected recommendation strengths, but the supplied summaries do not expose every full evidence profile or certainty rating. Where exact certainty was not extractable, ratings are conservative evidence-synthesis judgments based on the type of evidence summarized and should be verified against the full PAHO/WHO evidence tables before local adoption.

Sources: [PAHO/WHO guideline](#), [PAHO Digital Library](#), [PAHO news](#), [PMC](#).

B. WHO 2022 guideline: visceral leishmaniasis with HIV coinfection in East Africa and South-East Asia

Patient group	Recommendation	GRADE Evidence Quality	Strength
VL/HIV coinfection in East Africa	Use liposomal amphotericin B plus miltefosine rather than liposomal amphotericin B monotherapy. Regimen identified: liposomal amphotericin B up to 30 mg/kg total, 5 mg/kg on days 1, 3, 5, 7, 9, and 11, plus miltefosine 100 mg/day for 28 days.	Very Low	Conditional
VL/HIV coinfection in South-East Asia	Use liposomal amphotericin B plus miltefosine rather than liposomal amphotericin B monotherapy. Regimen identified: liposomal amphotericin B up to 30 mg/kg total, 5 mg/kg on days 1, 3, 5, 7, 9, and 11, plus miltefosine 100 mg/day for 14 days.	Very Low	Conditional

Patient group	Recommendation	GRADE Evidence Quality	Strength
VL/HIV coinfection after first VL episode, East Africa	Use secondary prophylaxis after the first VL episode , especially in patients at high relapse risk.	Very Low	Conditional
VL/HIV coinfection after first VL episode, South-East Asia	Use secondary prophylaxis after the first VL episode , especially in patients at high relapse risk.	Very Low	Conditional

High-risk relapse features include not being on antiretroviral therapy, CD4 count below 200 cells/mm³, multiple previous VL episodes, failure to achieve clinical or parasitological cure, no CD4 recovery at follow-up, and absence of secondary prophylaxis [WHO PDF](#), [NCBI Bookshelf](#), [NIH ClinicalInfo](#).

C. Diagnosis and management principles across WHO/PAHO sources

Area	Recommendation / principle	GRADE Evidence Quality	Strength
Suspected cutaneous leishmaniasis	Confirm diagnosis when feasible using parasitology, such as stained smear or culture from lesion material.	Low	Strong
Suspected mucocutaneous leishmaniasis	Use appropriate laboratory support, including serology where indicated, and assess mucosal involvement carefully.	Low	Strong
Molecular diagnosis	Use PCR where available because WHO/EMRO and later reviews report higher sensitivity than microscopy; recognize lack of assay standardization.	Moderate	Conditional
Visceral leishmaniasis	Treat promptly and completely because untreated VL can be fatal.	Moderate	Strong
Programme management	Integrate diagnosis and treatment with surveillance, vector/reservoir control where relevant, epidemic preparedness, medicine access, and community engagement.	Low	Strong

Sources: [EMRO manual PDF](#), [EMRO fact sheet PDF](#), [PAHO topic page](#), [Cutaneous Leishmaniasis 2022 review](#).

1.3 Guideline Development Methodology

Evidence Review Process

- PAHO/WHO's 2022 Americas guideline updated the 2013 recommendations because new evidence had accumulated and used **GRADE methodology** to evaluate evidence and develop recommendations [PAHO Digital Library](#).
- A PAHO Journal synthesis states that the guideline-development process considered **quality of evidence, balance of benefits and harms, patient values and preferences, applicability, costs, and implementation context**, and used the **DECIDE** approach for recommendation formulation [PMC](#).
- The WHO VL/HIV guideline followed the **WHO Handbook for guideline development**, used PICO questions, commissioned systematic review, applied **GRADE**, and used evidence-to-decision frameworks [NCBI Bookshelf](#).

Consensus Method

- WHO's VL/HIV guideline used a guideline development group with relevant expertise, geographic and gender representation, and online meetings from **28 September to 1 October 2020** [NCBI Bookshelf](#).
- The evidence-to-decision framework considered desirable and undesirable effects, certainty of evidence, values, resource implications, equity, acceptability, and feasibility [NCBI Bookshelf](#).

Conflict of Interest Management

- WHO reports that the VL/HIV guideline development group included experts with **no conflicts of interest** [NCBI Bookshelf](#).
- PAHO/WHO's guideline was developed under PAHO/WHO guideline-development standards and GRADE methodology [PAHO webinar page](#).

II. Evidence Summary After WHO Latest Guideline Release (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

Search Strategy

Evidence was synthesized from WHO/PAHO official guidance, WHO technical documents, PAHO guideline summaries, systematic reviews, network meta-analyses, randomized trials identified in secondary sources, diagnostic reviews, and authoritative clinical guidance.

Databases / Sources Searched

- WHO and PAHO official guideline and publication pages
- PAHO Digital Library / IRIS
- WHO IRIS and NCBI Bookshelf versions of WHO guideline materials
- PubMed / PMC-indexed reviews
- Peer-reviewed sources including *Frontiers in Cellular and Infection Microbiology* and *Nature Reviews Disease Primers* references
- NIH ClinicalInfo for opportunistic infection guidance
- IDSA/ASTMH clinical practice guideline as non-WHO contextual guidance

Inclusion Criteria

- Evidence relevant to diagnosis, treatment, secondary prophylaxis, or management of leishmaniasis.
- WHO/PAHO guidance and technical documents.
- Systematic reviews, meta-analyses, randomized trials, diagnostic reviews, and clinical practice guidance published or indexed after the 2022 PAHO/WHO second edition, where available.
- Evidence through **29 April 2026**.

Exclusion Criteria

- Non-clinical animal-only studies.
- Studies without extractable relevance to diagnosis, treatment, management, prevention, surveillance, or implementation.
- Non-WHO guidelines were not counted among the required five WHO/PAHO guideline documents, though they were considered as contextual evidence where clinically relevant.

2.2 Key New Studies

Study / source	Design	Key findings	GRADE Quality	Strength / implication
2025 systematic review and network meta-analysis of antileishmanial drugs	Systematic review and network meta-analysis including RCTs and observational studies through 21 February 2024	Amphotericin B may be the first-choice drug overall; pentavalent antimony may have the highest mortality during treatment; miltefosine had more gastrointestinal reactions; paromomycin had more injection-site pain and liver enzyme abnormalities. Authors emphasized need for more high-quality studies.	Low	Conditional support for amphotericin-based regimens; does not replace region-specific WHO/PAHO guidance
2023 multicountry Eastern Africa VL trial of paromomycin plus miltefosine	Randomized controlled multicountry trial	Evaluated paromomycin plus miltefosine as an alternative for visceral leishmaniasis in Eastern Africa; outcome details were not extractable from supplied summaries.	Moderate for design; Very Low for actionable update because outcomes not extractable	No change to WHO/PAHO recommendations without full results

Study / source	Design	Key findings	GRADE Quality	Strength / implication
2024 ChAd63-KH trial for post-kala-azar dermal leishmaniasis	Randomized double-blind phase IIb trial	Trial identified for PKDL treatment; efficacy and safety outcomes were not provided in supplied summaries.	Very Low for current decision-making	Research signal only
2026 systematic review of leishmaniasis recidivans	PRISMA systematic review, 43 papers from 17 countries, narrative synthesis	Leishmaniasis recidivans mainly affected children; microscopy and histology had poor accuracy; <i>L. tropica</i> commonly reported; trauma, immunosuppression, and irregular treatment were triggers; adherence was decisive; combination regimens may help persistence.	Low	Supports improved diagnostic confirmation, adherence support, and expert management
2022 diagnostic review of cutaneous leishmaniasis	Narrative review incorporating systematic review evidence	PCR has superior sensitivity and specificity compared with traditional diagnostic methods; qPCR may monitor response and predict treatment failure, but PCR protocols are not standardized.	Moderate	Conditional recommendation to use PCR where available
IDSA/ASTMH guideline contextual recommendation	Clinical practice guideline, non-WHO	Simple or healing cutaneous lesions in immunocompetent patients may not require treatment if not acquired in mucosal-risk regions; systemic therapy recommended for clinically manifest American mucosal leishmaniasis.	Low to Moderate depending question	Contextual; does not override WHO/PAHO guidance

Sources: [Frontiers 2025](#), [PMC/Frontiers review](#), [Nature Reviews Disease Primers](#), [PubMed 2026](#), [Cutaneous Leishmaniasis 2022 review](#), [IDSA/ASTMH guideline PDF](#).

2.3 Evidence Gaps and Future Research Needs

Evidence gap	Why it matters	Priority research need	GRADE certainty of current evidence
Species-specific treatment response in the Americas	Drug efficacy varies by <i>Leishmania</i> species and geography.	Larger RCTs stratified by species, region, lesion burden, mucosal risk, and immune status.	Low
Optimal treatment for immunocompromised patients	Immunocompromised patients have higher relapse and toxicity risks; PAHO recommends against pentavalent antimonials in this group.	Comparative trials of liposomal amphotericin B–based and combination regimens; relapse-prevention studies.	Low
Secondary prophylaxis for VL/HIV	WHO recommendations are conditional with very-low-certainty evidence.	RCTs or pragmatic studies comparing pentamidine, amphotericin formulations, and other prophylaxis strategies by region.	Very Low
PCR standardization	PCR is more sensitive than microscopy, but in-house assays limit comparability.	Standardized, field-adapted PCR/qPCR protocols and diagnostic accuracy studies.	Moderate for PCR accuracy; Low for implementation
Post-kala-azar dermal leishmaniasis	PKDL contributes to transmission and morbidity, especially in elimination settings.	Trials with clinical cure, relapse, infectivity, and safety outcomes.	Very Low
Leishmaniasis recidivans	Diagnosis is difficult and treatment evidence is fragmented.	Prospective registries and comparative treatment studies, especially in children.	Low
Youth and community engagement in leishmaniasis guidance	WHO has youth-engagement methods in other guideline areas, but leishmaniasis-specific youth engagement is not documented in supplied sources.	Include affected adolescents, young adults, caregivers, and endemic-community representatives in future guideline panels and implementation research.	Very Low

2.4 Updated Recommendations Based on New Evidence

Original Recommendation	New Evidence Impact	Updated Recommendation
PAHO/WHO 2022: use liposomal amphotericin B for visceral leishmaniasis where indicated.	2025 network meta-analysis supports amphotericin B as potentially preferred overall, consistent with amphotericin-based guidance, but evidence remains heterogeneous.	Maintain. Prefer liposomal amphotericin B when feasible, especially for severe disease, immunocompromised patients, and settings where toxicity of alternatives is a concern.
PAHO/WHO 2022: pentavalent antimonials and amphotericin B deoxycholate are conditional for VL.	Network meta-analysis raises safety concerns for pentavalent antimony; amphotericin B deoxycholate remains limited by toxicity compared with liposomal formulations.	Maintain conditional status. Use only when benefits outweigh harms, with toxicity monitoring and local susceptibility considerations.
PAHO/WHO 2022: avoid miltefosine for <i>L. infantum</i> VL.	No new evidence in supplied summaries overturns this; safety concerns and species-specific efficacy remain relevant.	Maintain strong recommendation against miltefosine for <i>L. infantum</i> VL.
PAHO/WHO 2022: intralesional antimonials strongly recommended for selected cutaneous leishmaniasis.	No post-guideline evidence in supplied summaries refutes this. Local therapy remains attractive for selected uncomplicated lesions.	Maintain. Use for appropriate localized cutaneous disease when expertise and follow-up are available.
PAHO/WHO 2022: ketoconazole removed from CL treatment options.	No new evidence supports reinstatement.	Maintain removal. Do not use ketoconazole for leishmaniasis treatment.
WHO 2022 VL/HIV: liposomal amphotericin B plus miltefosine conditionally recommended in East Africa and South-East Asia.	No supplied post-guideline outcome data supersede the WHO recommendation; recent trial activity supports ongoing evaluation of combinations.	Maintain conditional recommendation. Implement with ART integration, toxicity monitoring, pregnancy prevention for miltefosine, and relapse surveillance.
WHO 2022 VL/HIV: secondary prophylaxis after first VL episode conditionally recommended.	Evidence remains very low certainty, but relapse risk is clinically important.	Maintain conditional recommendation, especially for CD4 <200 cells/mm ³ , prior relapse, incomplete cure, or poor immune recovery.
WHO/EMRO: PCR more sensitive than microscopy.	2022 diagnostic review reinforces PCR superiority and qPCR utility, while highlighting lack of standardization.	Upgrade implementation emphasis. Use PCR/qPCR where available, but retain microscopy/culture where molecular

Original Recommendation	New Evidence Impact	Updated Recommendation
		testing is unavailable and pursue assay standardization.
Management of leishmaniasis recidivans not well defined in WHO/PAHO treatment tables.	2026 systematic review shows diagnostic difficulty, pediatric burden, persistence, and importance of adherence.	Add good-practice point. Confirm diagnosis with molecular methods where possible, assess immune status and prior adherence, and manage with expert consultation.

Practical Clinical Algorithm

1. Classify disease form

- Cutaneous, mucosal/mucocutaneous, visceral, post-kala-azar dermal leishmaniasis, or leishmaniasis recidivans.

2. Assess geography and likely species

- Treatment should be adapted to region and species because WHO/PAHO repeatedly emphasize that response varies by parasite species and geographic context [EMRO fact sheet PDF](#), [PAHO/WHO guideline](#).

3. Confirm diagnosis

- Use parasitology where feasible.
- Use PCR/qPCR when available, especially for atypical, mucosal, recurrent, or treatment-failure disease.

4. Treat urgently when visceral disease is suspected or confirmed

- VL can be fatal without treatment; all diagnosed VL patients require prompt, complete therapy [EMRO fact sheet PDF](#).

5. Select therapy by syndrome

- **Localized uncomplicated CL:** consider local therapy such as intralesional antimonials where appropriate.
- **Mucosal disease:** systemic therapy; PAHO/WHO supports pentavalent antimonials with or without pentoxifylline.
- **VL:** prefer liposomal amphotericin B where feasible; avoid miltefosine for *L. infantum* VL.
- **VL/HIV due to *L. donovani* in East Africa or South-East Asia:** use WHO-recommended liposomal amphotericin B plus miltefosine regimen and integrate ART and relapse prevention.

6. Monitor safety

- Antimonials: cardiac, pancreatic, hepatic, and renal toxicity.
- Amphotericin B formulations: renal function and electrolytes.
- Miltefosine: gastrointestinal toxicity, hepatic/renal monitoring, and strict pregnancy prevention.
- Paromomycin: injection-site reactions and liver enzymes.

7. Integrate public-health measures

- Surveillance, vector control, reservoir control where relevant, medicine access, community education, and outbreak preparedness remain essential [PAHO topic page](#), [EMRO fact sheet PDF](#).
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Overall Conclusion

As of **29 April 2026**, the most current WHO/PAHO leishmaniasis treatment guideline identified is PAHO/WHO's **2022 second edition** for the Americas. Its key actionable updates are removal of ketoconazole, stronger support for intralesional antimonials in selected cutaneous leishmaniasis, expanded species-specific support for miltefosine in cutaneous disease, strong support for pentavalent antimonials with or without pentoxifylline in mucosal disease, preference for amphotericin-based visceral leishmaniasis treatment, conditional status for more toxic older systemic agents, avoidance of miltefosine for *L. infantum* visceral leishmaniasis, and avoidance of pentavalent antimonials in immunocompromised patients.

Post-guideline evidence through **29 April 2026** broadly supports the direction of WHO/PAHO recommendations—especially amphotericin-based therapy and molecular diagnostics—but does not provide sufficiently definitive new evidence to replace the 2022 WHO/PAHO recommendations. The strongest needed updates are operational: better species-specific trials, standardized PCR/qPCR diagnostics, safer regimens for immunocompromised patients, stronger evidence for VL/HIV secondary prophylaxis, and structured inclusion of affected communities, including young people, in future guideline development.